

Assignment 4 - Computational neuroscience

Note: Claude commits to fix git complications and errors.

1. Introduction and Study Type

Purpose

Understanding how the human brain processes natural sounds requires bridging sensory physics with cognitive semantics. The **Superior Temporal Gyrus (STG)** acts as a crucial computational hub within the auditory cortex, bridging early acoustic feature extraction and higher-order semantic processing. This study uses **Representational Similarity Analysis (RSA)** to evaluate how well artificial neural networks emulate STG neural representations.

Dataset

The primary training data utilized is the **ESC-50** environmental sound dataset, consisting of 1,600 training samples, which was used to train all our models including the baseline and provided predefined models which are explained below

Models

We evaluate three custom architectures alongside a pre-trained baseline for this study:

Custom architectures:

- **WaveformModel:** A raw waveform baseline using multi-layer perceptrons with Tanh activations (approx 8.89M parameters) assuming no built-in auditory structure.
- **UninspiredModel:** A spectrogram-based model using 2D convolutional filters over frequency and time (approx 43K parameters) to loosely mimic tonotopic filtering.
- **InspiredModel:** A Convolutional Recurrent Neural Network (CRNN) using spectrogram inputs, a Gated Recurrent Unit (GRU), and ReLU6 activations (~620K parameters) to integrate information over time and cap biological firing rates.

Pre-trained benchmark:

- **YAMNet:** A large pre-trained audio event classification model serving as an architectural benchmark.

Study Type Classification

The study performed follows a **Contrastive and Constructive** framework:

- **Contrastive:** It compares different architectural hypotheses against biological brain data, such as raw waveforms versus spectrograms, and feedforward versus recurrent networks.

- **Constructive:** It implements and evaluates computational models in order to put theories of auditory processing into practice.

2. Methods

Representational Similarity Analysis (RSA) Pipeline

We performed RSA using the following standardized pipeline:

1. Extracting neural activations from STG recordings using `SantoroDataset()` and model layers via `extract_activations()`.
2. Constructing Representational Dissimilarity Matrices (RDMs) using Euclidean distances.
3. Comparing model RDMs to brain RDMs using Pearson correlation on the upper triangle, while also discussing alternative metrics like Spearman rank correlation.

Statistical Rigor & Confidence Intervals

To account for variance in training, we utilized the helper method `mean_ci` to compute 95% confidence intervals around the mean across multiple model seeds/runs ($n=5$ for waveform and inspired models, $n=5$ for uninspired models). Furthermore, **paired and Welch t-tests with Holm correction** were applied to evaluate statistically significant differences across layers and training conditions.

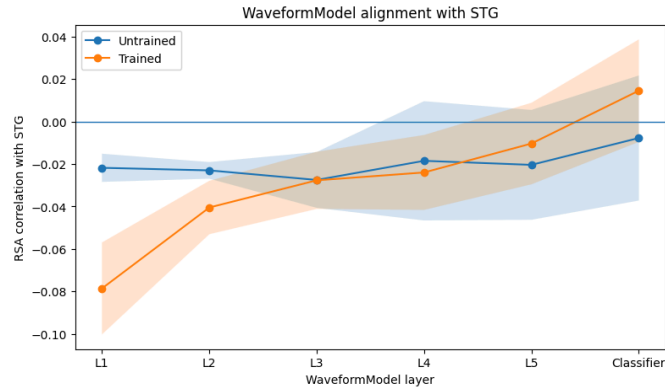
Visualization via t-SNE

We applied t-SNE to per-layer model activations to visualize how effectively different network depths cluster sound categories. This allowed us to compare learned latent geometries directly against neural recordings and understand how category separation evolves across network hierarchies.

3. Results

Training Effect: Untrained vs. Trained Models

- **Untrained Models:** Untrained baseline models which were initialized with random weights showed not very meaningful depth-dependent trends, maintaining a low random correlation with the STG data (r values hovering around -0.02). This indicates that the random initializations completely lack the biological representational structure.



- Trained Models:** Task optimization over 50 epochs on ESC-50 dataset demonstrated very clear upward alignment trends, confirming that the learning successfully shapes latent spaces toward the brain-like geometries. Training performance mirrored this convergence: the waveform model stalled at a chance level (~5% validation accuracy), the uninspired model reached ~30% validation accuracy, and the inspired model achieved ~40% validation accuracy (~60% train).

Model Architecture	Training Accuracy	Validation Accuracy
WaveformModel	—	~5% (chance level)
UninspiredModel	—	~30%
InspiredModel	~60%	~40%

Architectural Choices & Model Comparison

- **WaveformModel:** Peaked weakly at the final classifier layer ($r = 0.0145$), but its 95% confidence interval spanned zero ($[-0.0096, 0.0386]$), indicating minimal robust alignment.
- **UninspiredModel & InspiredModel:** Incorporating spectrogram inputs and convolutional filtering drastically improved alignment. The uninspired model peaked at Layer 2 ($r = 0.1009$ [$0.0877, 0.1141$]), while the inspired model achieved an even higher peak at Layer 2 ($r = 0.1134$ [$0.1028, 0.1239$]).
- **YAMNet Baseline:** Peaked strongly at intermediate Layer layer07relu ($r = 0.1151$) before dropping off significantly at the final embedding layer ($r = -0.0085$).

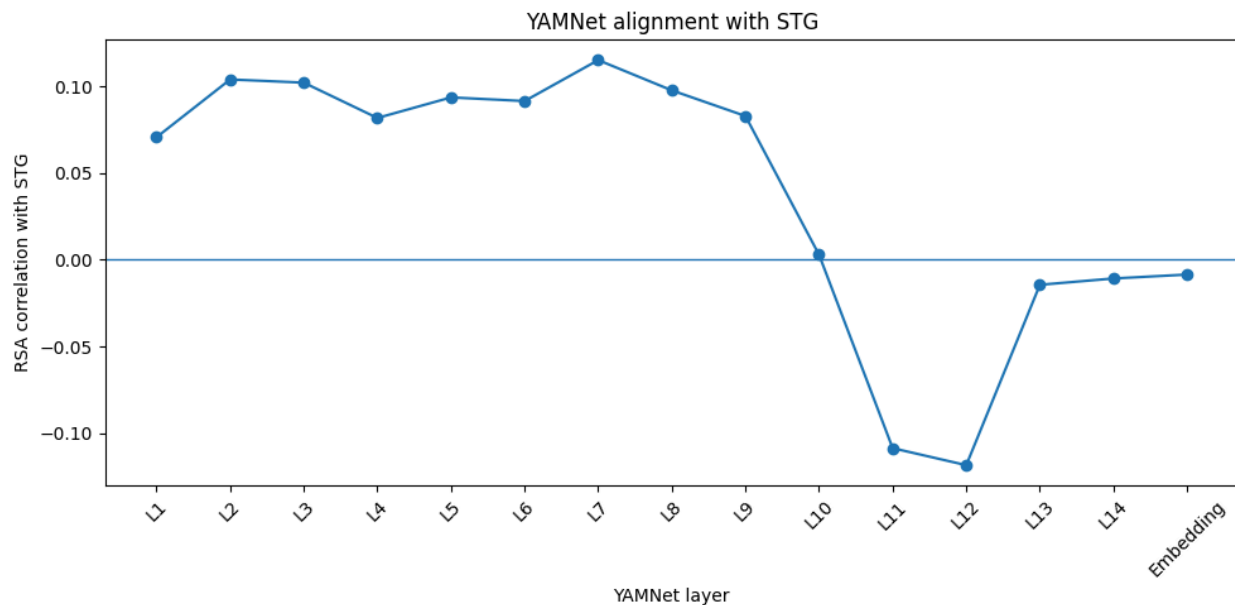


Figure 1. YAMNet-STG alignment per layer. Alignment peaks at intermediate layer L7 ($r = 0.1151$) and falls to near zero at the embedding ($r = -0.0085$).

Role of Model Depth and t-SNE Clustering

Across architectures, early-to-intermediate layers (relative depth 0.2 – 0.4) captured low-level acoustic features and spectrotemporal dynamics mirroring mid-level auditory cortex and STG data. Conversely, final task-specific layers often overfitted classification objectives, degrading brain-like representational geometry. t-SNE scatter plots confirmed that models with higher RSA alignment developed category clustering structures that closely mirrored biological organization.

Comparison of RSA Distance and Correlation Metrics

The choice of distance and correlation metrics noticeably influenced absolute alignment scores and shifted peak layers:

- **Euclidean Distances and Pearson Correlation:** Yielded higher absolute correlation values and clearer performance separation. Under this pair, the inspired model peaked at Layer 2 ($r = 0.1134$), closely matching YAMNet (layer07relu, $r = 0.1151$), while the waveform baseline peaked weakly at Layer 6 ($r = 0.0145$, CI crossing zero).
- **Correlation Distances and Spearman Rank Correlation:** Produced lower overall scores and shifted peak alignment layers deeper into the network. Peak values dropped across all architectures (waveform L4: $r = -0.0288$; untrained L5: $r = 0.0330$; inspired L5: $r = 0.0547$; YAMNet layer12relu: $r = 0.0980$). Euclidean distances paired with Pearson correlation thus provided a more sensitive scale for detecting structural representation.

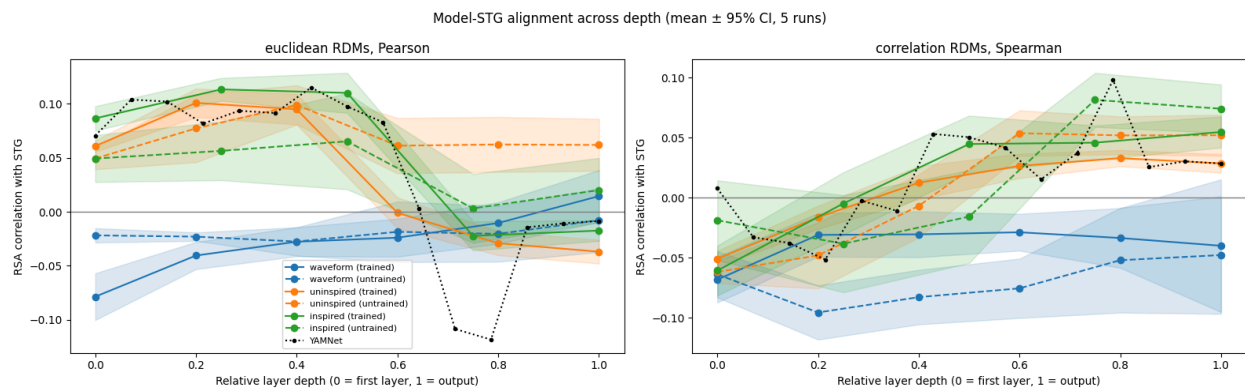


Figure 2. Model-STG alignment across normalised layer depth (0 = first layer, 1 = output). Left: Euclidean RDMs, Pearson. Right: correlation-distance RDMs, Spearman.

4. Discussion

Untrained vs. Trained Dynamics

Task-driven optimization is strictly necessary to shape artificial latent spaces so they mirror biological processing hierarchies. Without training or proper inductive biases, neural networks remain structurally unaligned with cortical data.

Why Architecture Matters

Spectrotemporal representations—specifically spectrogram inputs combined with convolutional filters mimicking tonotopy—bridge the gap between raw physical sound waves and the linguistic/semantic computations performed in the STG.

The Depth Paradox (YAMNet & Intermediate Peaks)

Intermediate layers outperform final classification layers in RSA due to a fundamental hierarchical divergence: the biological STG maintains a rich, high-dimensional sensory representation to handle complex acoustic environments, whereas artificial classification heads compress variance into discrete label boundaries, sacrificing fine-grained representational geometry.

5. Future Work & Neuroanatomical Refinement

To overcome current limitations (such as data scarcity in the 1,600-sample ESC-50 dataset) and improve brain alignment, future model iterations should incorporate advanced neuroanatomical features:

- **Multiscale Temporal Processing Streams:** Implementing parallel processing loops handling fast temporal fine-structure versus slow spectral modulations.
- **Recurrent Feedback Connections:** Adding top-down recurrent connections mimicking cortico-cortical feedback and predictive coding in the auditory hierarchy.
- **Self-Supervised Pretraining:** Utilizing unsupervised or self-supervised pretraining objectives (e.g., masked spectrogram modeling on large unlabeled audio corpora) prior to fine-tuning on downstream tasks.